Asadullah Ahmad

ICPC 2026 Silver Medalist
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Education

BS Computer Science, Punjab University College of Information Technology (PUCIT) 2023 – Present
CGPA: 3.6

Skills

Languages: C, C++, Java, Python, JavaScript, Kotlin
Web Technologies: HTML, CSS, React, Spring Boot, FastAPI, Flask, REST API
DevOps & Messaging: Docker, Kubernetes, RabbitMQ, Redis, Celery
Databases: SQL, MongoDB, Supabase, Firebase, Qdrant, Vector Databases, Graph Databases
AI/ML: LangChain, LangGraph, LLMs, NLP, RAG, Agentic AI, Prompt Engineering
Tools & Platforms: Git, Grep, LangFuse, LogFire, Google Cloud, Figma, Jira
Competitive Programming: LeetCode 650+, Strong DSA & Algorithms

Experience

AI/ML Intern Jun 2026 – Aug 2026
NetSol Technologies, Lahore

- Built an AI-powered chatbot using **LangChain** and **LangGraph** for multi-step conversational workflows.
- Fine-tuned a **YOLOX-S** model for signature detection, cutting the company's verification work from **7 days to 2 seconds**.
- Developed a **financial ratios extraction agent** with LangGraph, pulling cash, debt & cash-flow metrics at **100% accuracy** and reducing analysis time from **4 days to 10 seconds**.
- Containerized and deployed AI services with **Docker** on **Google Cloud Platform**, collaborating cross-functionally to ship features to production.

Teaching Assistant Jan 2025 – May 2026
PUCIT, Lahore

- **DSA** (Jul 2025–May 2026): Mentored students in DSA, competitive programming, and problem solving.
- **OOP** (Sep 2025–Jan 2026): Guided students in OOP concepts, design principles, and project evaluations.
- **Linear Algebra** (Jan 2025–May 2025): Supported students in vectors, matrices, and mathematical foundations.

Projects

Pixora GitHub | Live Demo
Built an image-to-search multimodal RAG platform with FastAPI and Next.js, using Google Gemini Vision for image understanding, Pinecone for vector search, and Tavily for real-time web-augmented streaming answers.

Lumio GitHub | Live Demo
Full-stack AI learning platform (Next.js, FastAPI, MongoDB, Gemini API) with JWT/Google OAuth, AI-generated adaptive quizzes, real-time leaderboards, analytics dashboards, and role-based access control.

BaatKaro GitHub | Live Demo
Production-grade multimodal AI assistant (Next.js, FastAPI, LangChain, Docker, MongoDB Atlas, GCP Cloud Run) with RAG, Face ID auth, streaming chat, speech-to-text/TTS, FAISS search, and Gemini+Groq fallback.

Signature Detector – YOLOX-S GitHub | Live Demo
Lightweight, production-ready signature detection API powered by a fine-tuned YOLOX-S model; converts PyTorch weights to ONNX for CPU-only inference (<50ms/page), fully containerized into a ∼200MB Docker image.

Financial Ratios Extractor GitHub | Live Demo
Financial analysis API extracting key ratios from company financial-statement PDFs (up to 35 pages), combining GREP for text extraction, Qdrant for retrieval, and Python for computations to generate accurate structured financial metrics.

Shanakht GitHub | Live Demo
Built an ID card information extraction system using OCR models to automatically detect and extract personal data fields from identity card images.

Hybrid RAG System GitHub
Hybrid RAG pipeline combining Neo4j knowledge graphs with dense vector databases, featuring hybrid retrieval, cross-encoder reranking, and context-aware LLM response generation.

Huffman Encoding Visualizer GitHub
Interactive Python tool visualizing Huffman tree construction, frequency analysis, and lossless compression with step-by-step animation.

Certifications

- ICPC 2026 Silver Medalist